Supplementary Material for *Large-scale traffic flow reconstruction from heterogeneous mobility sensors*

Federico Bellisardi, Chiara Mizzi, Alessandro Fabbri and Armando Bazzani

This supplementary material documents the meaning of the data layers, model variables, data-assimilation logic, validation metrics and webserver architecture supporting the main manuscript. Numerical choices are reported with their interpretation: some follow from the temporal resolution of the observations, some stabilize the simulator, and one (the 40 km separation of the example links) only serves the figures. The scripts that produce the revised numbers are listed in Section S8.

S1 Data Layers

The reconstruction combines four observational layers and one derived road support. Their roles are distinct and are not interchangeable (Table S1). The mobile-phone layer is the only one that describes regional demand at population scale, but it is too coarse to say which road was used. The road graph provides the missing physical support for route assignment. Coils provide accurate local counts, but only where a sensor exists. Viasat is spatially richer than coils, but it is a probe sample rather than a total-count system. The reconstructed link-flow table combines these partial views under the model assumptions described below.

Table S1: Meaning and role of the data layers.

Layer	Spatial/temporal support	Meaning in the workflow
Mobile-phone OD activity	Municipality pairs, hourly profiles	Population-scale demand. It defines where trips are generated and how demand changes over the day, but it does not identify the road sequence used by each trip.
Road graph	Directed polylines and nodes	Physical support on which demand is assigned. It stores connectivity, directionality, segment length and turn constraints.
MTS inductive loop detectors (coils)	Fixed points, 15-min counts	Local direct traffic measurements. They anchor the simulation and drive the data assimilation where observed counts are available.
Viasat GPS telemetry	Matched road links, hourly counts	Independent validation layer. Because it samples only part of the vehicle population, it is compared with simulation after estimating link-specific penetration.
Derived link-flow tables	Directed road links, hourly counts	Simulation output after optional data assimilation. These tables are the operational link-level reconstruction used for maps, validation and webserver delivery.

S1.1 Demand Structure

The weekday and weekend OD matrices are highly concentrated. This motivates using the observed spatial demand instead of injecting uniform trips across the region. The total-flow values in Table S2 are the activity weights entering the demand generator after aggregation over the selected January 2020 period. They are relative mobility mass, not vehicle counts. Positive cells count municipality pairs with non-zero activity. The Gini coefficient and top-share columns quantify concentration: values close to 0.8 and a top-1% share above 0.4 mean that a small number of OD relations dominate the signal. The model therefore uses the observed high-demand centres explicitly and assigns weakly observed endpoints to the random classes.

Table S2: Structure of the weekday and weekend (Saturday–Sunday) OD matrices.

Period	Total flow	Positive cells	Gini	Top 10% share	Top 1% share
Weekdays	2,665,186	87,212	0.788	0.708	0.448
Weekends	2,198,516	87,447	0.777	0.690	0.432

The ranking in Table S3 is used only to describe the heterogeneity of municipal activity and to identify the dominant urban centres. The last-ranked municipalities have mean activity equal to one because they are at the residual tail of the aggregated demand table. They are kept in the descriptive statistics, but the simulator does not assume that such low-activity entries carry enough information to define precise origin–destination routes on their own.

Table S3: Municipality activity ranking used to define high-demand centres.

Highest activity			Lowest activity		
Rank	Municipality	Mean points	Rank	Municipality	Mean points
1	Bologna	345,267	519	Montecalvo in Foglia	1
2	Modena	176,166	520	Porto San Giorgio	1
3	Parma	158,440	521	Trento	1
4	Ravenna	147,772	522	Montepulciano	1
5	Rimini	144,696	523	Vescovato	1
6	Reggio nell'Emilia	139,849	524	Tortona	1
7	Ferrara	103,067	525	Aviano	1
8	Piacenza	92,678	526	Malborghetto Valbruna	1
9	Forlì	86,078	527	Bolzano	1
10	Cesena	78,582	528	Grottammare	1

S1.2 Sensor and Telemetry Coverage

Table S4 reports data volume in terms of records, variables and physical interpretation. The 528 municipalities define the areal units of the demand layer. The 292 traffic stations are sparse compared with the road graph, so the assimilation is local and directional. The 5,760 rows of the 2020 coil table correspond to 60 days at 15-min resolution; each day contributes 96 samples per direction. The Viasat campaigns contain hundreds of thousands of hourly link observations, but the 4,186 matched road identifiers are still only a subset of the regional graph. This justifies using Viasat for temporal-profile validation and not as a complete observation layer.

S2 Simulation Runs and Source Classes

The final validation archive contains sixteen daily simulations, eight in February 2020 and eight in June–July 2021. The two periods were selected because independent Viasat validation campaigns are

Table S4: Observed data volumes used by the paper workflow.

Layer	Records	Variables	Interpretation
Municipality activity table	528	5	Ranked source and destination activity over municipalities.
Coil registry	292	5	Geographical catalogue of traffic-count stations.
Coil-to-road matching	292	9	Association between each traffic station and a graph polyline.
February–March 2020 coil table	5,760	496	Fifteen-minute directional traffic counts used for the coil-282 comparison and data-assimilation support.
Viasat 2020 campaign	803,904	10	Hourly GPS link-flow observations over 4,186 matched road identifiers.
Viasat 2021 campaign	703,416	10	Hourly GPS link-flow observations over the same matched road universe.

available for those dates. Viasat is not used to configure or assimilate the simulations; it provides the held-out probe layer against which simulated link profiles are compared. The two weeks do not span the seasonal variability of traffic.

Every run uses a 10 s integration time step and writes hourly outputs. The 10 s step is short enough to move agents through short links without creating coarse jumps at intersections, while remaining computationally light for regional daily runs. Hourly output is used because both the mobile-phone demand profiles and the validation figures are expressed at hourly scale; it also reduces noise relative to 15-min sensor counts while preserving morning, midday and evening traffic structure. The source count changes by day because the OD generator activates only demand components present in the selected daily schedule.

Table S5: Archived daily simulation configurations, summarized by meaning rather than file name.

Date	Sources	Output support
2020-02-10	758	Hourly directed links
2020-02-11	740	Hourly directed links
2020-02-12	762	Hourly directed links
2020-02-13	736	Hourly directed links
2020-02-14	722	Hourly directed links
2020-02-15	758	Hourly directed links
2020-02-16	980	Hourly directed links
2020-02-17	758	Hourly directed links
2021-06-28	763	Hourly directed links
2021-06-29	744	Hourly directed links
2021-06-30	766	Hourly directed links
2021-07-01	740	Hourly directed links
2021-07-02	726	Hourly directed links
2021-07-03	763	Hourly directed links
2021-07-04	983	Hourly directed links
2021-07-05	763	Hourly directed links

The slightly larger source counts on Sundays arise from the way residual weekend mobility is represented:

when the observed OD structure is less dominated by commuting pairs, more source components are needed to preserve regional coverage. This does not mean that Sunday traffic is necessarily larger than weekday traffic; it means that the demand generator uses a larger set of active source definitions to represent a more diffuse mobility pattern.

Table S6: Source-class counts across the archived simulations.

Class	Min/day	Max/day	Total	Endpoint logic
Fixed municipality OD	563	804	9,843	Observed source and observed destination.
Fixed province OD	55	78	1,003	Province-level source and destination.
Source-random	50	50	800	Observed source, sampled destination.
Random-destination	50	50	800	Sampled source, observed destination.
Random-random	1	1	16	Both endpoints sampled from the modelled region.

The fixed municipality OD class dominates because it is the part of the demand where both endpoints are directly observed. The two one-sided random classes are kept at 50 sources per day so that residual mobility can enter and leave the main observed centres without overwhelming the empirical OD matrix. The single random–random source is a low-intensity background component: it prevents the model from suppressing trips with neither endpoint in the dominant observed set, but it is deliberately small so that it cannot create an artificial uniform traffic field.

S3 `minimocas` Model Details

`minimocas` is an agent-based road simulator. Vehicles are represented by agents that move on directed polylines, choose a next node at junctions and accumulate link-level counts. The model state is summarized in Table S7. Agents are individual vehicles, without lane changing or car-following dynamics; the output of interest is the directional link-by-time flow field.

Table S7: Main vehicle-agent state variables.

State variable	Type/unit	Role
Identifier	integer	Distinguishes each simulated vehicle.
Status	categorical	Tracks whether the vehicle is active, completed, waiting, queued or interacting with transport logic.
Current road link	polyline id	Link occupied by the vehicle at the current time step.
Position on link	m	Longitudinal coordinate along the occupied link.
Signed speed	m s^{-1}	Magnitude gives speed; sign encodes direction.
Destination	node id	Current graph node pursued by the routing rule.
Previous and next node	node ids	Junction state used to move between links.
Route-deviation parameter	probability	Controls whether the vehicle follows or departs from the precomputed best path.
Distance and time counters	m, s	Accumulate travelled distance, trip time and lifetime.

Best paths are computed on a weighted graph with a Dijkstra-style engine. In the archived validation runs the routing weight reduces to polyline length, while forbidden turns are still excluded. The movement

update also contains a density-dependent speed rule:

$$v_p^{\text{target}} = \begin{cases} v_{\text{crit}}, & \rho_p > \rho_{\text{crit}}, \\ (1 - \rho_p/\rho_{\text{crit}})v_p, & \rho_p \le \rho_{\text{crit}}, \end{cases} \qquad v_p^{\text{target}} \ge v_{\text{crit}}. \tag{1}$$

The validation configuration uses $\rho_{\text{crit}} = 3/4.20$ m^{-1} and $v_{\text{crit}} = 10/3.6$ m s^{-1}.

The critical density corresponds to a compact but still interpretable spacing: roughly three vehicles over 4.2 m in the internal density convention used by the simulator. Its role is not to calibrate a detailed fundamental diagram; it prevents unrealistically high speeds when many agents occupy the same link. The critical speed is 10 km h^{-1}, converted to SI units, and acts as a lower operational speed in congested conditions. The rule should therefore be read as a stabilizing traffic-flow constraint rather than as a fitted car-following law.

Table S8: Numerical parameters and switches in the archived validation runs.

Parameter	Value	Role
Integration step	10 s	Numerical simulation step.
Output sampling	3600 s	Temporal resolution of saved link counts.
Source activation	30 s	Interval at which new vehicles are generated.
State grid cell	5000 m	Spatial scale of gridded state output.
Cartographic index cell	10000 m	Spatial index used for graph lookup.
Length routing weight	1.0	Makes routing cost proportional to length.
Speed routing weight	0.0	Disables speed-dependent routing in archived runs.
Initial speed range	15–25 m s^{-1}	Sampling range for generated vehicles.
Random-endpoint time budget	5600 s	Maximum time field used by random source classes.
Unique-link output	active	Saves the directional link counts used in the paper.

The values in Table S8 reflect a compromise between temporal fidelity and reproducibility. A 10 s integration step is small relative to the one-hour saved output and keeps route progression smooth. The 3600 s sampling interval is chosen because the OD demand and Viasat validation are analysed hourly; a finer output would mostly add sensor noise. The 30 s source-activation interval spreads injected agents over each hour instead of creating a single burst at the hour boundary. The 5000 m state grid and 10000 m cartographic index are lookup scales used to organize regional graph operations; they are much larger than individual road links, so they do not define the resolution of the final traffic field. Setting the length routing weight to 1.0 and the speed routing weight to 0.0 makes route choice depend on path length in the archived runs. This choice avoids mixing route assignment with uncertain speed calibration when the main objective is to validate link-flow profiles. The 15–25 m s^{-1} initial-speed range covers typical extra-urban speeds before density updates. The 5600 s random-endpoint time budget is a soft bound used when sampling residual trip endpoints: it allows trips longer than a one-hour commute but avoids generating unrealistic cross-region random journeys as the dominant residual class.

S4 Coil-based Data Assimilation

The data-assimilation step adjusts simulated unique-link counts using MTS coil residuals. Directional coil columns are mapped to cartographic `IN`/`OUT` directions. The archived validation runs were assimilated with the version of the code dated 30 March 2023; rerunning that version on the stored inputs reproduces the archived 10 February 2020 table to machine precision. In that version, gaps in an observed coil table are filled by linear interpolation in time, and when no observed table covers the simulated day (the 2021 runs) the weekday-specific seasonal average of each coil is used. The operational code now in the webserver fills missing values with seasonal weekday averages that combine the most recent year and earlier years with inverse standard-deviation weights.

For a coil-associated directed link i, let $u_i(h)$ be the unique-link count, $b_i(h)$ the simulated virtual-barrier count and $c_i(h)$ the measured coil count. The data-assimilation step estimates a barrier-to-unique

relation and a local relative residual,

$$r_i(h) = \frac{b_i(h)}{\max(u_i(h), 1)}, \qquad \Delta_i(h) = \frac{c_i(h) r_i(h) - b_i(h)}{\max(b_i(h), 1)}. \tag{2}$$

This residual becomes the additive increment $\delta_i(h) = u_i(h)\Delta_i(h)$ on the local directed link, so that the assimilated count equals $c_i(h)$ whenever $u_i, b_i \geq 1$, and is propagated to neighbouring links according to simulated turning shares. The ratio $r_i(h)$ is needed because a virtual barrier crossing and a unique-link occupancy count are not the same measurement. The former counts passages through a point-like section, whereas the latter counts distinct agents present on the whole directed polyline during the output interval. Using $r_i(h)$ converts the coil residual into the same scale as the simulated barrier before it is applied to the unique-link table. The denominator thresholds $\max(u_i(h), 1)$ and $\max(b_i(h), 1)$ prevent division by zero in night-time or low-flow hours; they do not add vehicles to the output, but only regularize the residual calculation.

Table S9: Stopping and exclusion rules in recursive data assimilation.

Rule	Implementation
Minimum propagated increment	Stop if the absolute assimilation increment is below one vehicle.
Maximum graph depth	Archived runs: stop once more than three links have been visited from the same coil (the counter is shared by all branches). Webserver version: five links.
Other sensor links	Archived runs: stop at a node as soon as one of its outgoing links is a coil polyline. Webserver version: skip only that link.
Forbidden turns	Do not propagate through transitions forbidden in the road graph.
Closed or patched roads	Do not propagate through excluded polylines.
Negative counts	Clip negative assimilated counts to zero before delivery.

The one-vehicle stopping rule in Table S9 is a discreteness threshold: below one vehicle an increment has no meaning in an hourly count table. The depth limit keeps the correction local, so that a single coil cannot reshape a large part of the regional network, and stopping at other sensor links leaves each observed location under the control of its own counter. Two features of the implementation should be kept in mind when reading the results. In both versions the increment passed to the second and later outgoing branches of a node is multiplied by the weights of the branches already processed, so those branches receive less than their turning share. Furthermore, the MTS tables store UTC timestamps, which both versions compare with the local clock of the simulation without conversion; the effect of this offset on the validation is quantified in the main text.

S4.1 Leave-one-out test

The leave-one-out test uses the assimilation code of the archived runs and the raw simulations of 10–17 February 2020. For every day and every coil with all 96 quarter-hours observed in both directions, the coil is removed from the list of assimilated coils and from the list of polylines at which the propagation stops, so that increments from neighbouring coils can reach its link. The assimilation is then rerun and the hourly counts $\tilde{u}(h)$ on the held-out link are compared with the hourly coil counts $c(h)$, in the same clock convention used by the assimilation. The error of a coil is $\sum |\tilde{u}(h) - c(h)| / \sum c(h)$ over hours, directions and days. Table S10 and Fig. S1 summarize the results. When the coil is kept in the assimilation the same error is 1.2%, the residual coming from hours with no simulated vehicle or barrier crossing on the link, where the increment does not reproduce $c(h)$. Of the 76 coils reached by the propagation, the median error goes from 49.4% to 44.7%.

Coil 282 is used in the main manuscript because both directions have complete 15-min coverage for the selected day. The daily sums are similar but not identical, as expected for a road with different dominant directions in the morning and in the afternoon. The opposite peak hours, 17:00 in direction 0 and 07:00 in direction 1, are consistent with a road segment that carries different dominant directions in the morning and afternoon.

Table S10: Leave-one-out test of the coil assimilation, 10–17 February 2020. For each coil with complete data on a given day, the coil is removed from the assimilation and from the set of polylines that stop the propagation; the error is $\sum_h |u(h) - c(h)| / \sum_h c(h)$ on the held-out coil link, summed over directions and days.

	Raw simulation	After assimilation
Pooled relative error [%]	59.8	56.7
Median over coils [%]	46.2	44.3
Interquartile range [%]	33.2–62.2	32.3–62.0
Coils tested		149
Coil-days		610
Coils reached by the propagation [%]		51
Coils improved / worsened [%]		39 / 13

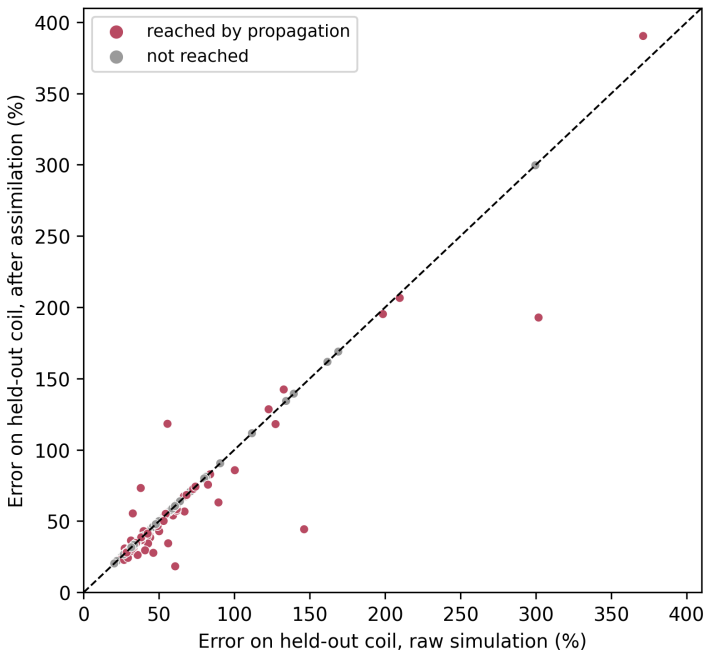


Figure S1: Leave-one-out test: error on each held-out coil link before (raw simulation) and after assimilating the other coils, 10–17 February 2020. Red: coils whose link receives an increment from the propagation; grey: coils not reached.

Table S11: Observed coil 282 data on Monday 10 February 2020 (local time).

Direction	Fifteen-minute samples	Daily sum	Peak hourly count	Peak hour
0	96	11,538	1,166	17:00
1	96	10,397	1,143	07:00

S5 Validation Tables

Viasat telemetry is compared with the reconstruction after link-specific penetration scaling (main text, Eqs. 2–5). The penetration η_i is the ratio between the Viasat probes and the simulated vehicles summed over the same days, so it is a daily sampling fraction of the road. It is estimated link by link because probe coverage depends on road type, fleet composition and local travel patterns. Table S12 shows that the 2021 campaign has a slightly lower penetration than the 2020 one; the Kolmogorov–Smirnov test rejects equal distributions at the 5% level, so the two campaigns are analysed separately. Penetration classes are defined by rounding $\log_{10}\eta_i$ to the nearest integer, because η_i spans four orders of magnitude; their shares are shown in Fig. 7 of the main text and their profile errors in Table 1 of the main text.

Table S12: Link penetration η_i (Viasat probes over simulated vehicles on the same days). Two-sample Kolmogorov–Smirnov test between the campaigns: $D = 0.072$, $p = 2.3 \times 10^{-2}$.

Campaign	Links	Median $\log_{10}\eta$	Median η [%]	Mean η [%]
2020	848	-2.299	0.502	0.785
2021	845	-2.361	0.435	0.670

The candidate set for the example links is the upper class of a three-class k-means partition of the standardized $\log_{10}\eta_i$, which in 2020 contains the 487 links with $\eta_i \geq 0.40\%$. Before clustering, each simulated profile is divided by its daily total, so that Ward linkage acts on the 24 hourly shares and not on volumes. On raw counts the Ward distance is dominated by volume: the three raw-count clusters used in the earlier version of this analysis (74, 112 and 301 links) are disjoint bands of daily traffic, 1,001–6,384, 6,597–15,446 and 14,638–55,856 simulated vehicles per day. The partition on shares has an adjusted Rand index of 0.18 with the raw-count partition, so the two groupings are largely different. The dendrogram is cut into three clusters, at Ward distance 0.15; three is also the number of example links that fit in one validation figure. Figure S2 shows the mean hourly shares of the three subclusters.

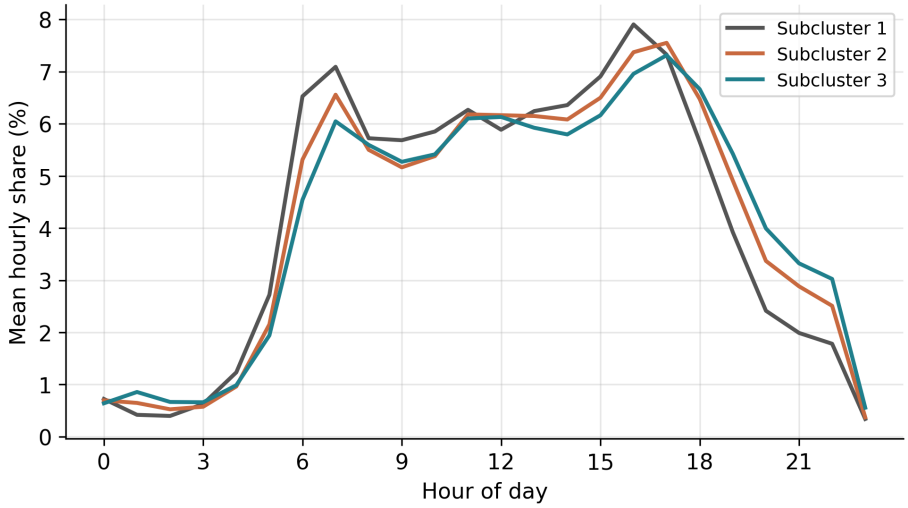


Figure S2: Mean normalized simulated profile of the three subclusters of Fig. 9 of the main text (51, 78 and 358 links).

The selection of the example links in Table S14 follows three rules. The three roads belong to different subclusters. Polyline 104671 is fixed because its position near Forlimpopoli gives it a direct interpretation. In the other two subclusters, among the links with at least 300 simulated vehicles per day and 50 Viasat probes in the week (all 487 candidates), we take the pair with the lowest total MAE whose pairwise distances, including the distance to 104671, are at least 40 km. The two selected links are thus among the best of the candidate set, as the last column of Table S14 shows; the full distribution is in Table 1 and Fig. 8 of the main text.

Table S13: Candidate-filtering logic for the representative validation links.

Selection stage	Links	Purpose
Common simulated–Viasat links in 2020	930	Initial matched comparison set.
Usable hourly profiles	848	Non-zero simulated and GPS totals.
Upper k-means class of $\log_{10}\eta_i$	487	Candidates for subclustering, $\eta_i \geq 0.40\%$.
Displayed representative links	3	One illustrative link per temporal subcluster.

Table S14: Representative polylines of Fig. 10 in the main text. The last column is the percentage of the 848 usable 2020 links with a lower MAE. Minimum pairwise separation: 45.0 km.

Subcluster	Polyline	MAE [pp]	ε [%]	Links with lower MAE [%]
1	104671	1.22	29	40
2	134419	0.81	19	5
3	64544	0.72	17	1

S6 Aggregate Road-safety Context

Panel **a** reports the yearly numbers of accidents and deaths in 2022–2024: about 16,700 accidents per year, with 16,679, 16,808 and 16,757 in the three years. Panel **b** shows the road environment: urban roads account for most accidents in every year, motorways and ramps for a small fraction. Panel **c** compares the seasonal shares of accidents with those of the simulated weekday exposure. All exposure runs use the same January 2020 OD matrices, so the seasonal differences of the simulated exposure do not come from the demand, and the comparison with the accident seasonality is only descriptive. **[TODO: state whether the 2023 exposure runs were assimilated with the 2023 coil data]**

S7 Webserver Architecture

The project also implements the reconstruction as an operational web system. The webserver wraps the configuration generator, the simulator and the assimilation into a service that users launch from a browser; the modelling equations are unchanged, except for the assimilation rules listed in Table S9.

The separation between interface, simulation service and worker process keeps long simulations from blocking the browser. The graphical interface stores metadata and displays results; the backend validates requests and coordinates simulation and data assimilation; the worker performs the computationally expensive network loading. The scheduled updater is included because the counter layer is not static: new MTS observations must be converted into the standardized format before they can be used for monitoring or data assimilation.

The workflow in Table S16 mirrors the scientific pipeline of the main manuscript. User input defines the temporal window, the backend generates a model configuration, the simulator produces raw flows, the data-assimilation step applies coil information, and the final link-flow table is stored for visualization and export. A regional simulation takes longer than a normal HTTP request, so the interface polls the run status.

The conceptual API in Table S17 exposes only operations needed by the workflow. The road-graph export allows the browser to draw the same network used by the simulator. The result query applies data assimilation at delivery time so the downloaded table and visualized map correspond to the data-assimilated reconstruction, not merely to the raw simulation.

S8 Reproducibility Notes

The numbers of the revised manuscript are produced by the scripts in `article/python/revision/` of the project repository: `validation_all_links.py` (validation on all links, baselines, penetration

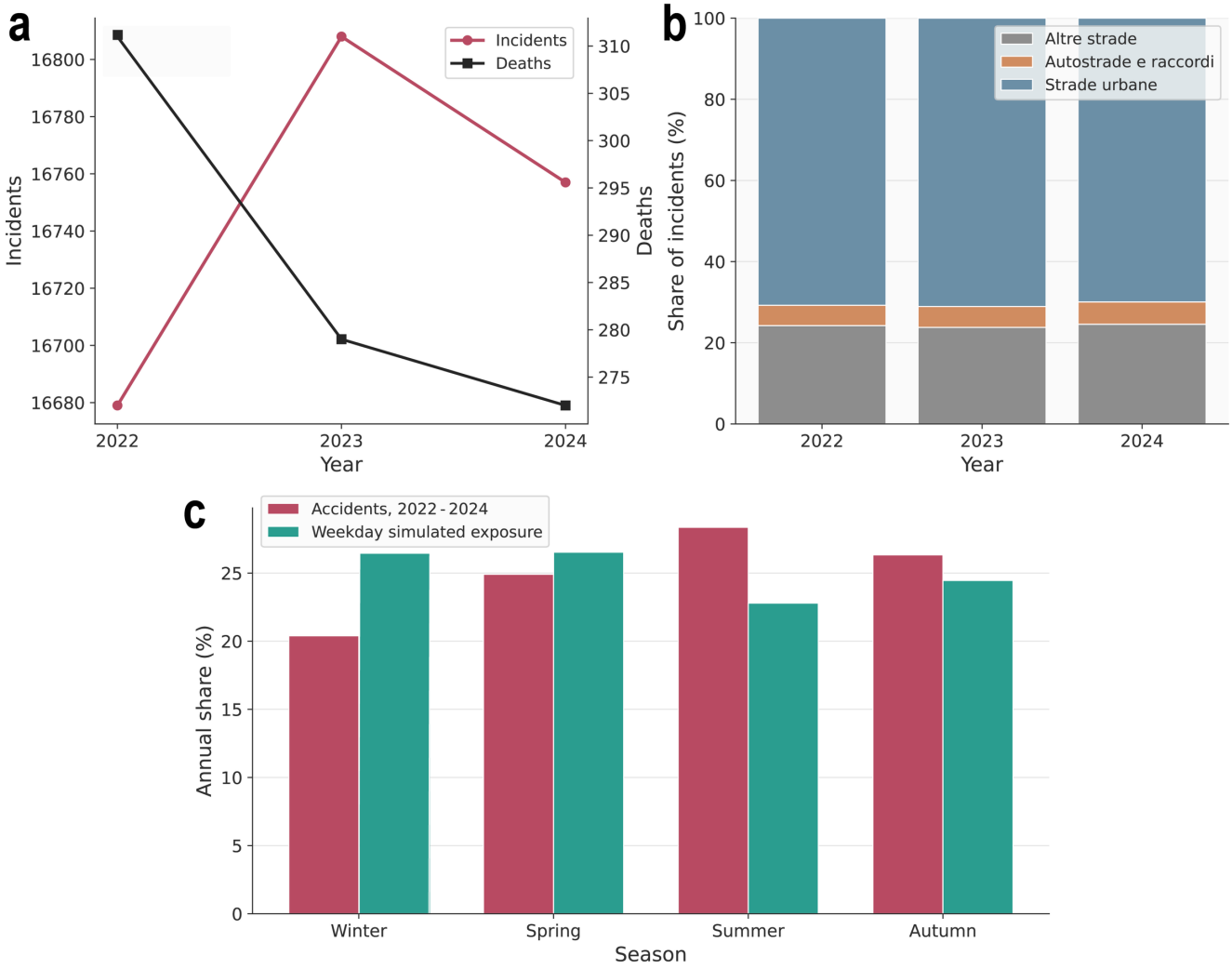


Figure S3: Accident data used in the exposure analysis of the main text: (a) yearly accidents and deaths, (b) road environment, (c) seasonal shares of accidents and of simulated weekday exposure.

Table S15: Logical components of the webserver.

Component	Function
Django graphical interface	Authenticated browser interface. It allows a user to submit a simulation interval, inspect submitted runs, download results and open interactive maps.
FastAPI simulation service	Programmatic backend that receives simulation requests, generates a run configuration, launches the simulator in a separate process and exposes status/results endpoints.
Configuration generator	Converts requested dates, coil availability and OD activity into the simulation input used by the backend.
Simulation worker	Runs `minimocas` from the generated configuration and writes status updates while the model advances.
Data-assimilation module	Applies the coil-driven data-assimilation step before results are returned to the user. A special mode returns provincial min–max intervals for institutional users.
Interactive map layer	Renders cartography and simulated fluxes by colouring road polylines by quantile or by selected result values.
MTS monitoring views	Display coil time series, station maps, rankings, clusters and local anomaly diagnostics.
Scheduled data updater	Downloads and converts the public MTS traffic records into the standardized coil tables consumed by the simulator.
Reverse proxy and containers	Provide a deployable service layout with separate GUI, simulation API, scheduler and web proxy containers.

Table S16: End-user workflow implemented by the dashboard.

Step	Action	Result
1	Choose interval	The browser submits start time, stop time and output sampling resolution.
2	Create simulation	The backend validates the request, generates a unique run identifier and builds the configuration.
3	Run asynchronously	The simulation starts in a separate process while the interface polls progress.
4	Assimilate results	After completion, the backend applies the default coil-based data-assimilation step or an institutional min–max adjustment.
5	Persist metadata	The interface stores run identifier, user, status, runtime, completion and result location in the application database.
6	Visualize and export	Users can inspect time profiles, view flux maps and download data-assimilated link-flow tables.

Table S17: Conceptual API exposed by the simulation backend.

Operation	Meaning
Welcome/status handshake	Returns service availability and version.
Simulation submission	Accepts a time interval and creates a simulation task.
Road-graph export	Returns the road graph geometry used by the interface.
Progress query	Reports whether a submitted run is running, complete, invalid or in error.
Result query	When requested after completion, applies data assimilation and returns the reconstructed link-flow table as JSON or CSV.

classes, clustering and example links), `loo/run_loo.py` (leave-one-out test, with the March 2023 assimilation code and cartography extracted from the project history), `loo/tz_sensitivity.py` (coil clock), `incident_risk.py` (accident exposure and bootstrap of the exponent), `coil_282.py`, `make_tables.py` and `compose_figures.py`. They read the archived simulation outputs, the Viasat extracts and the MTS tables, and write summary files to `article/results/revision/`. No new traffic simulation was run; the assimilation was rerun only for the leave-one-out and clock tests. The mobile-phone and Viasat inputs are proprietary and are not in the repository.